

A Novel Transformer-Based Framework for Bangla Tense Classification

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Abstract—Tense classification in Bengali natural language processing is a challenging task due to the language’s complex grammatical structure. This study evaluates five transformer architectures, BGE-M3 Embedding, BanglaBERT, mBERT, miniLM-L12-v2, and XLM-RoBERTa, on the BanglaTense dataset, comprising 17,819 annotated sentences categorized into past, present, and future tenses. All transformer models achieved more than 99% accuracy, demonstrating their robustness and effectiveness for Bengali tense classification. BGE-M3 Embedding achieves state-of-the-art performance with 99.49% accuracy, 99.49% precision, 99.50% recall, and 99.50% F1-score, significantly surpassing previous benchmarks. The results highlight the importance of leveraging advanced learning paradigms, such as contrastive learning, for optimizing semantic similarity in tense classification tasks. These findings provide empirical evidence for the efficacy of modern transformer architectures in Bengali computational linguistics and establish new benchmarks for temporal classification tasks.

Index Terms—Bengali natural language processing, tense classification, transformer models, BGE-M3 Embedding, BERT, contrastive learning, temporal analysis

I. INTRODUCTION

Bengali, one of the most prominent Eastern Indo-Aryan languages, serves as the native language for approximately 272.8 million speakers worldwide, making it the 7th most spoken language globally. The language’s complex grammatical structure, particularly its intricate tense system, presents significant challenges for natural language processing (NLP) applications, including machine translation, text summarization, and temporal information extraction. The computational complexity of Bengali tense classification extends beyond simple morphological analysis. Bengali employs sophisticated verb conjugation patterns, temporal markers, and contextual indicators that necessitate a deep understanding of semantics. Traditional rule-based approaches fail to capture the nuanced relationships between syntactic structures and temporal meanings, while classical machine learning methods struggle with the morphological richness and contextual dependencies inherent in Bengali text.

Recent advances in transformer architectures [1] have revolutionized natural language understanding tasks across multiple languages. However, systematic evaluation of these models for Bengali tense classification remains completely unexplored. While existing research has focused primarily on

sentiment analysis and document classification in Bengali, the specific challenge of temporal classification using transformer models has received no attention, creating a significant gap in the literature.

This paper introduces BGE-M3 Embedding [2] as the optimal solution for Bengali tense classification, providing the first comprehensive evaluation of transformer-based models for this task. Our proposed approach demonstrates superior performance over established baseline approaches on the BanglaTense dataset [3]. Our experimental results reveal that the proposed BGE-M3 Embedding model outperforms traditional masked language modeling architectures, providing novel insights for tense classification tasks.

The key contributions of this work are:

- Proposing BGE-M3 Embedding as the optimal solution for Bengali tense classification.
- Comprehensive evaluation of five transformer architectures for Bengali tense classification.
- Providing empirical evidence that contrastive learning models excel at temporal understanding tasks.
- Demonstrating that both language-specific and multilingual models achieve excellent performance on well-defined temporal classification challenges.
- Establishing new performance benchmarks that advance the state-of-the-art for the Bengali NLP community.

II. RELATED WORK

A. Dataset Development

The foundation for Bengali tense classification research was established by Bijoy et al. [3], who introduced the BanglaTense dataset containing 17,819 annotated Bengali sentences categorized into past, present, and future tenses. This dataset was systematically collected from diverse sources, including blogs, Facebook pages, magazines, and news articles, ensuring balanced representation across temporal categories with validation by native Bengali speakers. Earlier work by Ayman et al. [4] had developed a smaller version of this corpus with 13,500 sentences, which served as the preliminary foundation for subsequent research.

B. Ensemble Methods

Ayman et al. [4] pioneered ensemble approaches for Bengali tense classification using their 13,500-sentence corpus. Their stacked ensemble model combined five base models (Random Forest, SVM, XGBoost, LSTM, GRU) through a neural network meta-learner, achieving 85% accuracy. Individual model performance varied significantly: Random Forest (78.89%), LSTM (77.48%), GRU (77.11%), SVM (75.11%), XGBoost (74.52%), RNN (70.67%), Multinomial Naive Bayes (67.04%), and K-Nearest Neighbors (56.52%). The ensemble approach demonstrated a 6.11% improvement over the best individual model.

C. Deep Learning Baselines

Bijoy et al. [3] established comprehensive deep learning baselines on the expanded BanglaTense dataset. BiLSTM achieved the highest performance at 95.92% accuracy, followed by Conv1D-BiLSTM (95.76%), Conv1D-LSTM (95.61%), and LSTM (95.51%). Notably, Conv1D-LSTM demonstrated exceptional future tense classification with 98.50% accuracy, highlighting the importance of architectural choices for different temporal categories.

D. State-of-the-Art RNN Architectures

Parvez et al. [5] conducted a comprehensive RNN evaluation on the BanglaTense dataset. GRU emerged as the optimal architecture with 95.59% accuracy, outperforming Bi-GRU (95.37%), Bi-LSTM (95.17%), and LSTM (95.01%). All models exceeded 95% accuracy, establishing strong baselines. GRU's success was attributed to computational efficiency and superior sequential modeling capabilities for temporal patterns.

E. Research Gap: Transformer Models

Despite significant progress with ensemble and recurrent approaches, systematic evaluation of transformer models for Bengali tense classification remains completely unexplored. This represents a critical gap, as transformer architectures have revolutionized NLP through attention mechanisms and deep contextual representations. The current state-of-the-art performance of 95.92% achieved by Bijoy et al. [3] establishes a strong baseline, but the potential for transformer models suggests opportunities for advancement through contrastive learning approaches optimized for semantic similarity rather than token prediction. Our work addresses this gap by providing the first comprehensive transformer evaluation for Bengali tense classification.

III. DATASET OVERVIEW

A. BanglaTense Dataset Characteristics

The BanglaTense dataset serves as the foundation for our experimental evaluation, providing a comprehensive collection of 17,819 manually annotated Bengali sentences categorized into past (5,629 sentences), present (6,101 sentences), and future (6,089 sentences). The dataset contains a balanced distribution across tense categories, ensuring fair and unbiased model training and evaluation.

B. Data Collection and Sources

The dataset was systematically collected from diverse Bengali sources, including publicly accessible blogs, Facebook pages, magazines, and news articles from Bangladesh. This multi-source approach ensures comprehensive coverage of both formal and informal linguistic styles, capturing the natural variability inherent in contemporary Bengali usage.

C. Data Quality and Validation

The dataset underwent rigorous quality control with three-step validation by native Bengali speakers. Annotation consistency was ensured through majority voting among linguistic experts, comprehensive preprocessing protocols, and balanced domain coverage across different text types and stylistic variations.

D. Dataset Statistics

Table I presents the distribution of dataset entries, word counts, and word percentages for each tense category in the BanglaTense dataset.

Tense	Entries	Words	Word %
Future	6,089	38,311	36.13%
Past	5,629	31,302	29.52%
Present	6,101	36,417	34.35%
Total	17,819	106,030	100.00%

TABLE I
DISTRIBUTION OF DATASET ENTRIES, WORD COUNTS, AND WORD PERCENTAGES FOR EACH TENSE

E. Dataset Examples

Figure 1 presents one example sentence for each tense category, illustrating the diversity of temporal annotations in the dataset.

Sample	Tense
আমরা ভালো সময়ের স্বপ্ন দেখছি	Present
আমার জয় আপনাদের সমর্থনের ফল হবে	Future
নিবন্ধন বৃদ্ধি পেয়েছিল	Past

Fig. 1. Examples of Bengali sentences categorized by tense

IV. METHODOLOGY

A. Model Selection and Architecture

Our experimental methodology encompasses systematic evaluation of five transformer-based architectures, representing diverse training paradigms and architectural innovations. The model selection strategy encompasses three primary categories: contrastive models, multilingual BERT [6] architectures, and language-specific BERT variants.

Contrastive Models:

- **BGE-M3 Embedding:** State-of-the-art multilingual contrastive model (568M parameters) specifically optimized for cross-lingual semantic understanding through contrastive learning methodologies.

- **miniLM-L12-v2**: Compact contrastive model (118M parameters) [7] optimized for semantic similarity tasks through knowledge distillation and contrastive learning protocols.

Multilingual BERT Models:

- **XLM-RoBERTa**: Cross-lingual RoBERTa model (279M parameters) [8] trained on 100 languages utilizing optimized training procedures.
- **mBERT**: Multilingual BERT model (179M parameters) [9] employing MLM and NSP objectives across diverse multilingual corpora.

Bengali-Specific Models:

- **BanglaBERT**: Enhanced Bengali BERT variant (110M parameters) [10] incorporating advanced training methodologies on specialized Bengali corpora used with specialized normalization [11].

B. Key Architectural Distinctions

Traditional BERT models utilize masked language modeling (MLM), optimizing for token prediction accuracy based on surrounding context. Contrastive learning represents a fundamentally different paradigm where models learn to encode semantically similar texts in proximate vector spaces while maximizing distance between dissimilar representations [12]. This methodology optimizes semantic similarity metrics rather than token prediction, making contrastive models particularly effective for tense classification tasks requiring subtle temporal distinction capabilities.

C. Experimental Setup

Data Preprocessing: All text inputs undergo standardized preprocessing, including tokenization using model-specific tokenizers (with maximum sequence length of 512), handling of special Bengali characters, and normalization of text encoding. We maintain the original text structure to preserve tense-specific linguistic features crucial for classification.

Data Split: The BanglaTense dataset, comprising 17,819 annotated sentences, is partitioned using a stratified 80:10:10 split ratio to ensure balanced representation across all tense categories. The training set contains 14,255 instances (80%), the validation set contains 1,782 instances (10%), and the test set contains 1,782 instances (10%). The stratified approach maintains proportional class distribution across all splits: approximately 32% Past tense, 34% Present tense, and 34% Future tense in each partition. This splitting strategy ensures that model evaluation is conducted on completely unseen data while providing sufficient training examples for robust model optimization.

Training and Evaluation Protocol: Models are trained exclusively on the training set, with hyperparameter optimization performed using the validation set. Final performance evaluation and all reported results are obtained from the test set to ensure unbiased assessment of model generalization capabilities. Early stopping is implemented based on validation set performance to prevent overfitting.

D. Custom Model Architecture Implementation

We designed a unified framework to standardize fine-tuning across transformer models, ensuring compatibility while preserving each model's unique features. All encoders are paired with a classification head for Bengali tense prediction.

Architecture: Tokenized Bengali text is processed by the encoder, with the [CLS] token (or first token for contrastive models) used for classification. The classification head applies dropout ($p=0.1$), followed by a linear layer and softmax activation to predict the tense category.

Training: Cross-entropy loss is used for end-to-end optimization. Dense embeddings from contrastive models are integrated with the same classification head to ensure fair comparison across architectures.

The workflow of the model architecture is illustrated in Figure 2.

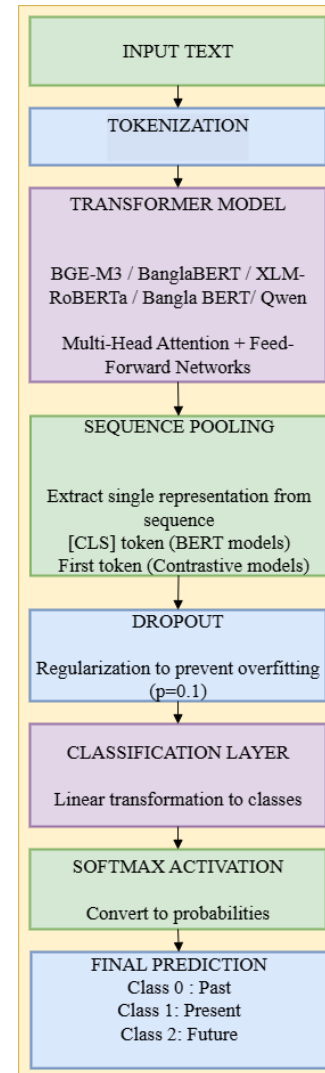


Fig. 2. Model architecture for Bengali tense classification.

E. Hyperparameter Optimization

We conduct systematic hyperparameter optimization to ensure fair comparison:

- **Learning Rate:** Evaluated rates include 1e-5, 2e-5, 3e-5, 4e-5, 5e-5.
- **Batch Size:** Three sizes evaluated (8, 16, 32).

F. Evaluation Metrics

We employ comprehensive evaluation metrics:

- **Accuracy:** Macro average classification accuracy across all tense categories, measuring the proportion of correctly classified instances for each class averaged equally.
- **Precision:** Macro average precision across all classes, evaluating the proportion of true positive predictions among all positive predictions.
- **Recall:** Macro average recall across all classes, assessing the proportion of true positive predictions among all actual positives.
- **F1-Score:** Harmonic mean of macro average precision and recall, providing a balanced measure of model performance.
- **AUC-ROC:** Area Under the Receiver Operating Characteristic Curve, measuring the model's ability to distinguish between classes across different thresholds.

G. Implementation Details

All experiments are implemented using PyTorch [13] and the Transformers [14] library on a high-performance workstation equipped with an NVIDIA GeForce RTX 5080 GPU, AMD Ryzen 9 7900X processor, and 64GB DDR5 RAM. The models are fine-tuned using the AdamW optimizer [15] with a learning rate scheduler for efficient convergence.

V. RESULTS AND ANALYSIS

A. Overall Performance Comparison

Table II summarizes the comprehensive performance evaluation of five transformer models on the test dataset, including accuracy, precision, recall, F1-score, and AUC scores.

TABLE II
OVERALL PERFORMANCE AND AUC COMPARISON OF TRANSFORMER MODELS

Model	Accuracy	Precision	Recall	F1-Score	AUC Score
BGE-M3 Embedding	99.49%	99.49%	99.50%	99.50%	99.90%
BanglaBERT	99.44%	99.43%	99.44%	99.43%	99.96%
XLNet	99.33%	99.33%	99.32%	99.32%	99.87%
miniLM-L12-v2	99.27%	99.27%	99.27%	99.27%	99.83%
mBERT	99.16%	99.17%	99.15%	99.16%	99.94%

All models exhibit exceptional performance across all evaluation metrics, including accuracy, precision, recall, F1-score, and AUC scores. Each model achieves AUC scores above 99.80%, demonstrating their strong discrimination capabilities. Among the evaluated models, BGE-M3 Embedding achieves the highest accuracy (99.49%), precision (99.49%), recall (99.50%), and F1-score (99.50%), establishing itself as the best-performing model. BanglaBERT, while slightly

behind in accuracy (99.44%), demonstrates competitive precision (99.43%), recall (99.44%), and F1-score (99.43%), and achieves the highest AUC score (99.96%), highlighting its superior ability to distinguish between classes across different thresholds.

XLNet and miniLM-L12-v2 also deliver consistent performance, with XLNet achieving 99.33% accuracy and miniLM-L12-v2 achieving 99.27% across all metrics. mBERT, while ranking lowest among the transformer models, still achieves strong results with 99.16% accuracy, 99.17% precision, 99.15% recall, and 99.16% F1-score, demonstrating the robustness of transformer architectures for Bengali tense classification tasks.

Top Performing Models: BGE-M3 Embedding sets a new benchmark for Bengali tense classification. BanglaBERT also shows strong results, highlighting the benefits of language-specific training. Consistent performance across all metrics confirms the reliability of transformer architectures for Bengali NLP.

B. Comparison with Previous Bengali Tense Classification Studies

Table III presents a comparative analysis of accuracy improvements over previous Bengali tense classification studies. The results demonstrate the substantial advancement achieved by transformer-based models, particularly BGE-M3 Embedding, compared to earlier ensemble and recurrent neural network approaches. Notably, previous works relied on ensemble and RNN architectures, achieving up to 95.92% accuracy, while our proposed transformer-based approach surpasses these results with 99.49% accuracy.

Study	Approach	Accuracy
Ayman et al. [4]	Stacked Ensemble	85.00%
Bijoy et al. [3]	BiLSTM	95.92%
Parvez et al. [5]	GRU	95.59%
Our proposed approach	BGE-M3 Embedding	99.49%

TABLE III
PERFORMANCE IMPROVEMENT OVER PREVIOUS BENGALI TENSE CLASSIFICATION STUDIES

C. Per-Class Performance Analysis Across All Models

Table IV presents detailed per-class performance metrics for all transformer models, revealing how each architecture handles different tense categories.

The per-class analysis reveals several important patterns across tense categories. Future tense consistently achieves the highest performance across all models, with BGE-M3 Embedding and BanglaBERT both achieving perfect recall (100%) for this category. This suggests that future tense markers in Bengali are more distinctive and easier for transformer models to identify.

Past tense demonstrates consistently high recall across all models (99.11% to 99.82%), indicating robust identification of past tense linguistic patterns. Present tense shows the most challenging classification pattern, with slightly lower

TABLE IV
PER-CLASS PERFORMANCE ANALYSIS - ALL MODELS

Model	Tense	Precision	Recall	F1-Score
BGE-M3 Embedding	Past	99.29%	99.82%	99.56%
	Present	99.83%	98.69%	99.26%
	Future	99.35%	100.00%	99.67%
BanglaBERT	Past	99.11%	99.64%	99.38%
	Present	99.67%	98.52%	99.09%
	Future	99.51%	100.00%	99.75%
XLM-RoBERTa	Past	98.93%	99.47%	99.20%
	Present	99.51%	98.36%	98.93%
	Future	99.34%	99.84%	99.59%
miniLM-L12-v2	Past	98.75%	99.29%	99.02%
	Present	99.34%	98.20%	98.77%
	Future	99.18%	99.67%	99.43%
mBERT	Past	98.57%	99.11%	98.84%
	Present	99.18%	98.03%	98.60%
	Future	99.01%	99.51%	99.26%

recall rates (98.03% to 98.69%) across all models, suggesting potential ambiguity in present tense contextual markers.

BGE-M3 Embedding maintains superior performance across all three categories, while BanglaBERT shows particularly strong future tense classification with the highest F1-score (99.75%) for this category. The consistent performance patterns across models validate the reliability of the BanglaTense dataset annotations and the effectiveness of transformer architectures for Bengali temporal classification.

D. Hyperparameter Analysis

Table V presents the optimal hyperparameters selected for all transformer models after systematic optimization, providing insights into the configuration requirements for achieving optimal performance.

TABLE V
OPTIMAL HYPERPARAMETERS FOR ALL TRANSFORMER MODELS

Model	Learning Rate	Batch Size	Epochs
BGE-M3 Embedding	3e-5	16	15
BanglaBERT	3e-5	16	16
XLM-RoBERTa	2e-5	32	18
miniLM-L12-v2	4e-5	16	29
mBERT	2e-5	16	11

The hyperparameter analysis reveals interesting patterns across different model architectures. Contrastive models (BGE-M3 Embedding and miniLM-L12-v2) show a preference for higher learning rates (3e-5 and 4e-5, respectively), while multilingual BERT models (XLM-RoBERTa and mBERT) perform optimally with lower learning rates (2e-5). The epoch requirements vary significantly, with mBERT converging fastest (11 epochs) and miniLM-L12-v2 requiring extended training (29 epochs) for optimal performance.

E. Confusion Matrix Analysis

Figure 3 presents the confusion matrix for the BGE-M3 Embedding model, illustrating its classification performance across the three tense categories: Past, Present, and Future.

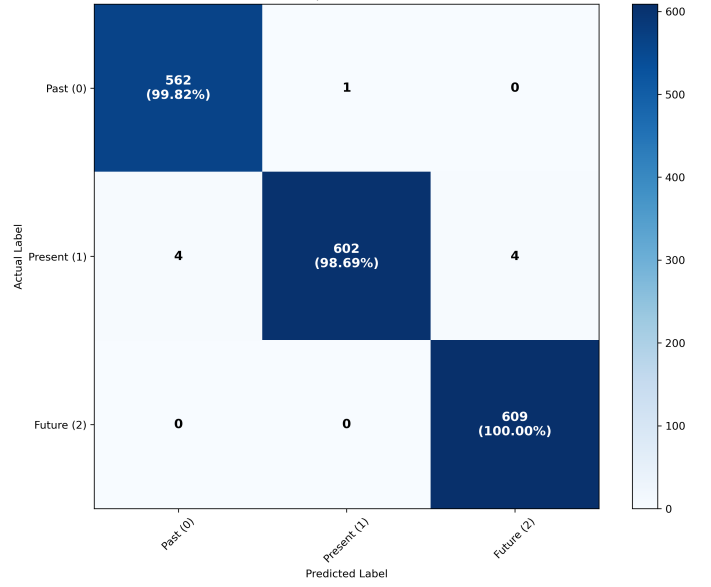


Fig. 3. Confusion Matrix for BGE-M3 Embedding Model showing classification results across Past (0), Present (1), and Future (2) tense categories.

The confusion matrix reveals the following key observations:

- **Past Tense:** Out of 563 instances, 562 were correctly classified, achieving 99.82% recall. Only 1 instance was misclassified as Present tense.
- **Present Tense:** Out of 610 instances, 602 were correctly classified, achieving 98.69% recall. The model misclassified 4 instances as Past tense and 4 as Future tense.
- **Future Tense:** All 609 instances were correctly classified, achieving perfect recall (100%).

The majority of misclassifications occur in the Present tense category, which is inherently more ambiguous due to overlapping temporal markers and contextual dependencies. Misclassifications primarily involve confusion with Past and Future tense categories.

These results highlight the robustness of the BGE-M3 Embedding model, particularly in Future tense classification, where it achieves perfect recall. The slight drop in recall for the Present tense reflects the inherent linguistic complexity of Bengali temporal markers, which often overlap across categories.

VI. DISCUSSION

The exceptional performance of our proposed BGE-M3 Embedding (99.49% accuracy) substantiates the theoretical advantages of contrastive learning for tense classification tasks. The significant improvement of 3.57% points over the previous state-of-the-art (95.92% by Bijoy et al.) demonstrates the effectiveness of our proposed approach. The contrastive learning paradigm enables these architectures to capture subtle semantic distinctions crucial for temporal classification, where multiple linguistic features contribute to tense determination.

The success of contrastive models can be attributed to their optimization for semantic similarity detection, which effectively captures temporal relationships in Bengali text. Unlike traditional masked language modeling that focuses on token prediction, contrastive learning optimizes for semantic understanding, making it inherently suitable for tasks requiring temporal distinction.

BanglaBERT (99.44%) achieves highly competitive performance with BGE-M3 Embedding (99.49%), with only a 0.05% difference, demonstrating that both language-specific and multilingual approaches are equally effective for Bengali tense classification. The strong performance of XLM-RoBERTa (99.33%) and other multilingual models suggests that cross-lingual semantic knowledge contributes significantly to temporal understanding [16]. This indicates that both approaches have equivalent merit for Bengali temporal classification tasks.

The results establish a clear performance hierarchy based on accuracy: BGE-M3 Embedding (99.49%) > BanglaBERT (99.44%) > XLM-RoBERTa (99.33%) > miniLM-L12-v2 (99.27%) > mBERT (99.16%). This ranking reveals important architectural insights for Bengali tense classification.

The remarkably narrow performance range of only 0.33% indicates that all transformer models achieve exceptional results on this task.

VII. CONCLUSION

This study provides the first comprehensive evaluation of transformer architectures for Bengali tense classification, demonstrating that BGE-M3 Embedding is the optimal model for this task. BGE-M3 Embedding achieves state-of-the-art results with 99.49% accuracy, surpassing previous baselines and outperforming traditional deep learning approaches by a margin of 3.57%. These findings highlight the effectiveness of contrastive learning and transformer-based methods for temporal classification in Bengali, setting new benchmarks for future research in Bengali natural language processing.

VIII. FUTURE WORK

While this study establishes strong benchmarks for Bengali tense classification, several avenues remain for future exploration. Further research could investigate model interpretability to better understand how transformer architectures capture temporal features in Bengali text. Expanding the dataset to include more diverse sources and informal language could improve generalizability.

IX. ACKNOWLEDGMENT

The authors gratefully acknowledge Bijoy et al. [3] for creating the BanglaTense dataset, which served as the foundation for this research.

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